

CSE 5004-Graph Theory and Algorithms

Exercise – 10

Calculate the structural metrics and the centrality metrics on `les_miserables_graph` that returns coappearance network of characters in the novel *Les Miserables*. Give top 10-important nodes for each centrality metric.

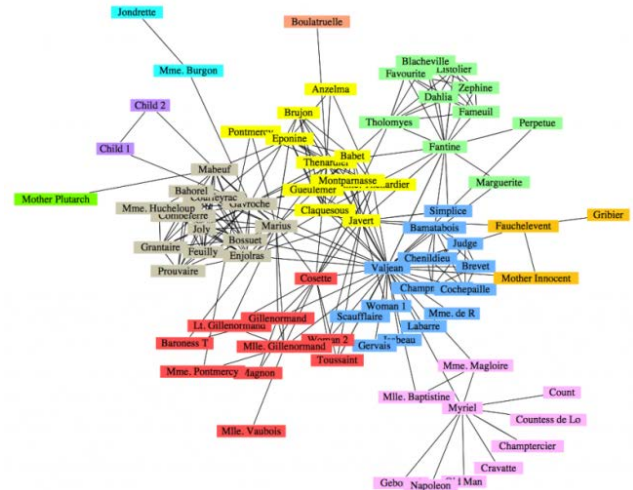
```
import networkx as nx
G = nx.les_miserables_graph()
```

Structural metrics

- Density
- Diameter

Centrality metrics

- Degree
- Betweenness
- Closeness
- Eigenvector
- Eccentricity



You can also experience the same metrics on a real-life social network from the following repositories or any other resources:

- <https://snap.stanford.edu/data/>
- <https://github.com/gephi/gephi/wiki/Datasets>